Path integral method

1. Path integral in quantum mechanics

1.1 Functional method

Functional S maps a function to numbers, i.e.

$$S : x(t) \to S[x(t)]. \tag{1.1}$$

As we all know, the Hamilton operator in 1-dim is $H = \frac{p^2}{2m} + V(x)$, now let us consider the amplitude for motion of a particle from x_a to x_b. For this, we introduce the evolution operator $U(x_a, x_b; t)$ in Heisenburg picture, so we have

$$U(x_a, x_b; T) = \langle x_b | \, e^{-\mathrm{i}HT/\hbar} \, | x_a \rangle \,, \tag{1.2}$$

where the Hamiltonian **is already quantized**, but in the following calculation, the Hamiltonian that in the path integral **is classical** due to the path integral itself is the process of quantization. **So we don't need to quantize anything inside of the integration**. So the path integral is directly defined from this operator. The basic idea is to cut the space and time into small pieces, and add them together, i.e.,

$$U(x_a, x_b; T) = \int \mathcal{D}x(t) e^{\mathrm{i}S[x(t)]/\hbar}, \tag{1.3}$$

where $S[x(t)]$ is the action that depends on the paths. In QM, we can expand it as

$$S[x(t)] = \int_0^T \mathrm{d}t \, L = \int_0^T \mathrm{d}t \left[\frac{1}{2}mv^2 - V(x)\right]. \tag{1.4}$$

And the integration measure $\mathcal{D}x(t)$ accounts for all possible paths from x_a to x_b, i.e.,

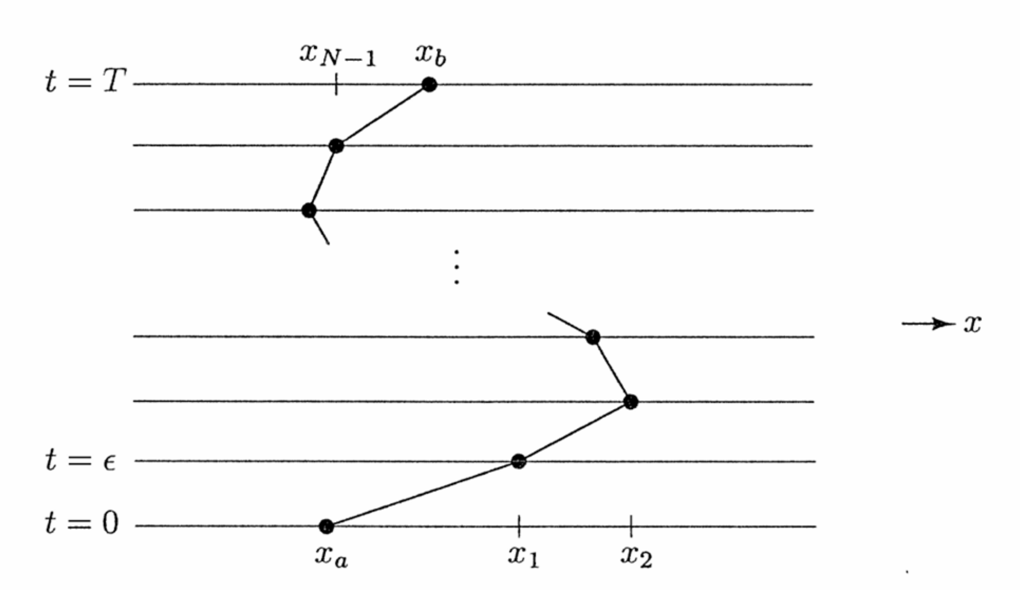
$$\sum_{\substack{\text{all paths from} \\ x_a \text{ to } x_b}} \longrightarrow \int \mathcal{D}x(t). \tag{1.5}$$

Now we need to prove that the definition we gave that can derive the same quantization result. Firstly, we discretize the time interval from 0 to T in steps $\Delta t = \varepsilon$:

$$S = \int_0^T \mathrm{d}t \left[\frac{1}{2}m\dot{x}^2 - V(x)\right] = \sum_k \left\{ \frac{m}{2} \frac{(x_{k+1} - x_k)^2}{\varepsilon} - \varepsilon V\left(\frac{x_{k+1} + x_k}{2}\right) \right\}. \tag{1.6}$$

The path integral build up from N discrete time intervals

$$\mathcal{D}x(t) \equiv \frac{1}{c(\varepsilon)} \int \frac{\mathrm{d}x_1}{c(\varepsilon)} \frac{\mathrm{d}x_2}{c(\varepsilon)} \ldots \frac{\mathrm{d}x_N}{c(\varepsilon)} = \frac{1}{c(\varepsilon)} \prod_{k=1}^{N} \int_{-\infty}^{+\infty} \frac{\mathrm{d}x_k}{c(\varepsilon)}, \tag{1.7}$$



where $\frac{1}{c(\varepsilon)}$ is an undetermined normalization factor. Now we consider the evolution operator $U(x_a, x_b; T)$ evolve from $T - \varepsilon$ to T in the interval $[x_a, x_b]$, which is the last step in (1.7):

$$\begin{aligned} U(x_a, x_b; T) &= \langle x_b | \, e^{-\mathrm{i}H[(T-\varepsilon)+\varepsilon]/\hbar} \, | x_a \rangle = \int \mathrm{d}x' \, \langle x_b | \, e^{-\mathrm{i}H\varepsilon/\hbar} \, | x' \rangle \, \langle x' | \, e^{-\mathrm{i}H(T-\varepsilon)/\hbar} \, | x_a \rangle \\ &= \int_{-\infty}^{+\infty} \frac{\mathrm{d}x'}{c(\varepsilon)} \exp\left[\frac{\mathrm{i}}{\hbar} \frac{m(x_b - x')^2}{2\varepsilon} - \frac{\mathrm{i}}{\hbar} \varepsilon V\left(\frac{x_b + x'}{2} \right) \right] U(x_a, x'; T - \varepsilon) \end{aligned} \tag{1.8}$$

Notice that we can expand $U(x_a, x'; T - \varepsilon)$ around x_b

$$U(x_a, x'; T - \varepsilon) = \left(1 + (x' - x_b) \frac{\partial}{\partial x_b} + \frac{1}{2} (x' - x_b)^2 \frac{\partial^2}{\partial x_b^2} + ... \right) U(x_a, x_b; T - \varepsilon) \tag{1.9}$$

and for $\varepsilon \to 0$, the potential term can also expand as

$$\exp\left\{ \frac{\mathrm{i}}{\hbar} \varepsilon V\left(\frac{x_b - x_{x'}}{2} \right) \right\} \overset{x' \to x_b}{=} e^{\frac{\mathrm{i}}{\hbar} \varepsilon V(x_b)} = 1 - \frac{i}{\hbar} \varepsilon V(x_b) + ... \tag{1.10}$$

Thus, (1.9) can be rewritten as

$$\begin{aligned} U(x_a, x_b; T) = &\int_{-\infty}^{+\infty} \frac{\mathrm{d}x'}{c(\varepsilon)} \exp\left\{ \frac{\mathrm{i}}{\hbar} \frac{m(x_b - x')^2}{\varepsilon} \right\} \left[1 - \frac{\mathrm{i}\varepsilon}{\hbar} V(x_b) + \ldots \right] \\ &\times \left[1 + (x' - x_b) \frac{\partial}{\partial x_b} + \frac{1}{2} (x' - x_b)^2 \frac{\partial^2}{\partial x_b^2} + \ldots \right] U(x_a, x_b; T - \varepsilon) \end{aligned} \tag{1.11}$$

To do this integration the Gauss integration formulas are very useful in the next:

$$\int_{-\infty}^{+\infty} \mathrm{d}\xi \, e^{-b\xi^2} = \sqrt{\frac{\pi}{b}}; \tag{1.12}$$

$$\int_{-\infty}^{+\infty} \mathrm{d}\xi \, \xi e^{-b\xi^2} = 0; \tag{1.13}$$

$$\int_{-\infty}^{+\infty} \mathrm{d}\xi \, \xi^2 e^{-b\xi^2} = \frac{1}{2b} \sqrt{\frac{\pi}{b}}. \tag{1.14}$$

Thus,

$$U(x_a, x_b; T) = \frac{1}{c(\varepsilon)} \sqrt{\frac{\pi}{b}} \left[1 - \frac{\mathrm{i}\varepsilon}{\hbar} V(x_b) + \frac{1}{4b} \frac{\partial^2}{\partial x_b^2} + \mathcal{O}(\varepsilon^2) \right] U(x_a, x_b; T - \varepsilon), \tag{1.15}$$

where $b = \frac{-im}{2\varepsilon\hbar}$. When $\varepsilon \to 0$,

$$U(x_a, x_b; T) = \overbrace{\frac{1}{c(\varepsilon)}\sqrt{\frac{\pi}{b}}}^{=1} U(x_a, x_b; T-\varepsilon)$$
$$\Rightarrow c(\varepsilon) = \sqrt{\frac{\pi}{b}}. \tag{1.16}$$

So we have the Schrödinger equation for the evolution operator

$$\mathrm{i}\hbar \frac{U(x_a, x_b; T) - U(x_a, x_b; T-\varepsilon)}{\varepsilon} = \left(V(x_b) - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_b^2}\right) U(x_a, x_b; T-\varepsilon). \tag{1.17}$$

So we have derived the familiar differential equation with the canonical quantization did. Now we also need to consider the boundary condition. For $T \to 0$, canonical quantization requires $U(x_a, x_b; 0) = \langle x_b| \mathbb{1} |x_a\rangle\, \delta(x_a - x_b)$. For path integral, we firstly take a piece of time slice, since $T \to 0$ there is no difference for labeling the space interval, so we can just take $\delta x = x_b - x_a$:

$$\lim_{\varepsilon \to 0} \frac{1}{c(\varepsilon)} \exp\left[\frac{im}{\hbar}\frac{(x_b - x_a)^2}{\varepsilon}\right] = \delta(x_b - x_a) \tag{1.18}$$

Thus,

$$\lim_{\varepsilon \to 0} \int \mathcal{D}x(t) e^{\mathrm{i}S[x(t)]/\hbar} = \lim_{\varepsilon \to 0} \frac{1}{c(\varepsilon)} \exp\left[\frac{im}{\hbar}\frac{(x_b - x_a)^2}{\varepsilon}\right] \to \delta(x_a - x_b) \tag{1.19}$$

- Path integral formulas quantize system through summation/integration over all possible paths; weighted by the phase $e^{-\mathrm{i}S[x(t)]/\hbar}$.
- In the limit $\hbar \to 0$ (classical physics) path integral is dominated by classical path with $\delta S = 0$.

1.2 Generalization to quantum system with coordinates $\{q_i\}$ and conjugate momenta $\{p_i\}$

Now we consider the general case for a quantum system with Hamiltonian $H(q, p)$, and evolution operator $U(q_a, q_b; T) = \langle q_b| e^{-\mathrm{i}HT} |q_a\rangle$. Split up $f(q, p) = e^{-\mathrm{i}HT}$ in N time intervals $\Delta t = \varepsilon$

$$e^{-\mathrm{i}HT} = \prod_{k=1}^{N} e^{-\mathrm{i}H\varepsilon} = e^{-\mathrm{i}H\varepsilon}\mathbb{1}e^{-\mathrm{i}H\varepsilon}\mathbb{1}\cdots\mathbb{1}e^{-\mathrm{i}H\varepsilon}, \tag{1.20}$$

with $\mathbb{1} = \left(\prod_i \int \mathrm{d}q_k^i\right) |q_k\rangle \langle q_k|$. At k-th position,

$$\langle q_k + 1| f(q) |q_k\rangle \xrightarrow{\varepsilon = 0} \langle q_k + 1| 1 - i\varepsilon H + \mathcal{O}(\varepsilon^2) |q_k\rangle \tag{1.21}$$

For a function that its variable is only q, we have

$$\langle q_{k+1}| f(q) |q_k\rangle = f(q_k) \prod_i \delta(q_k^i - q_{k+1}^i) = f\left(\frac{q_k + q_{k+1}}{2}\right) \left(\prod_i \int \frac{\mathrm{d}p_i}{2\pi}\right) \exp\left\{\mathrm{i}\sum_i p_k^i (q_{k+1}^i - q_k^i)\right\}, \tag{1.22}$$

and for a function that its variable is only p, we have

$$\langle q_{k+1}| f(p) |q_k\rangle = \left(\prod_i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) f(p_k) \exp\left\{\mathrm{i}\sum_i p_k^i (q_{k+1}^i - q_k^i)\right\}. \tag{1.23}$$

In general, we normally face the more complicated case that using the Weyl ordering is convenient to keep the properties we have in the above, i.e.,

$$\langle q_{k+1}| \frac{1}{4}\left(q^2p^2 + 2qp^2q + p^2q^2\right)|q_k\rangle = \left(\frac{q_{k+1}+q_k}{2}\right)^2 \langle q_k + 1|\, p^2\, |q_k\rangle\,, \tag{1.24}$$

where the Weyl order is defined as

$$W(p,q) = \frac{1}{2}\{x,p\}. \tag{1.25}$$

Assume $H(p,q)$ is Weyl oerdered, i.e.,

$$\langle q_{k+1}|\, H(p,q)\, |q_k\rangle = \left(\prod_i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) H\left(\frac{q_{k+1}+q_k}{2}, p_k\right) \exp\left\{\mathrm{i}\sum_i p_k^i(q_{k+1}^i - q_k^i)\right\}. \tag{1.26}$$

Thus

$$\langle q_{k+1}|\, e^{-\mathrm{i}H\varepsilon}\, |q_k\rangle = \left(\prod_i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) \exp\left\{-\mathrm{i}\varepsilon H\left(\frac{q_{k+1}+q_k}{2}, p_k\right)\right\} \exp\left\{\mathrm{i}\sum_i p_k^i(q_{k+1}^i - q_k^i)\right\}. \tag{1.27}$$

Take $q_0 = q_a$ and $q_N = q_b$, the integration are

$$U(q_0, q_N; T) = \left(\prod_{i,k} \int \mathrm{d}q_k^i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) \exp\left\{\mathrm{i}\sum_k \left[\sum_i p_k^i(q_{k+1}^i - q_k^i) - \varepsilon H\left(\frac{q_{k+1}+q_k}{2}, p_k\right)\right]\right\}. \tag{1.28}$$

In path integral notation

$$U(q_0, q_N; T) = \left(\prod_{i,k} \int \mathcal{D}q^i(t) \int \mathcal{D}p^i(t)\right) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \left[\sum_i p^i\dot{q}^i - H(p,q)\right]\right\}. \tag{1.29}$$

2. Path integral in the Scalar field

For a scalar field $\phi(\vec{x},t)$, we can simply change the variables $\vec{q}_i(t) \longmapsto \phi(\vec{x},t)$, the evolution operator is then became:

$$U(\phi_a, \phi_b; T) = \langle \phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle. \tag{1.30}$$

The action S and Hamiltonian H for thw scalar field are defined as:

$$S = \int \mathrm{d}^4x \left(\frac{1}{2}(\partial\cdot\phi)^2 - V(\phi)\right), \tag{1.31}$$

$$H = \int \mathrm{d}^3x \left(\frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi)\right). \tag{1.32}$$

The path integral form can then be simply seen from (1.29) as

$$\langle \phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \int_{\phi(0,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x) \int \mathcal{D}\pi(x) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left(\pi\dot{\phi} - \frac{1}{2}\pi^2 - \frac{1}{2}(\nabla\phi)^2 - V(\phi)\right)\right\} \tag{1.33}$$

$$= \int \mathcal{D}\phi(x) \int \mathcal{D}(\pi(x) - \dot{\phi}) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[-\frac{1}{2}\left(-2\pi\dot{\phi} + \pi^2 + \dot{\phi}^2 - \dot{\phi}^2\right) - \frac{1}{2}(\nabla\phi)^2 - V(\phi)\right]\right\} \tag{1.34}$$

To cancel the $\int \mathcal{D}\pi(x)$, we change the variable $\mathcal{D}\pi(x) \to \mathcal{D}(\pi(x) - \partial_0\phi)$, i.e.,

$$\langle\phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \int \mathcal{D}\phi(x) \int \mathcal{D}(\pi(x) - \dot{\phi}) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[-\frac{1}{2}\left(\pi - \dot{\phi}\right)^2 + \frac{1}{2}(\partial\phi)^2 - V(\phi)\right]\right\} \tag{1.35}$$

$$= \int \mathcal{D}\phi(x) \int \mathcal{D}\tilde{\pi}(x) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[-\frac{1}{2}\tilde{\pi}(x)^2 + \frac{1}{2}(\partial\phi)^2 - V(\phi)\right]\right\}. \tag{1.36}$$

Thus

$$\langle\phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \mathcal{C} \int \mathcal{D}\phi(x) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[\frac{1}{2}(\partial\phi)^2 - V(\phi)\right]\right\}, \tag{1.37}$$

where

$$\mathcal{C} = \int \mathcal{D}\tilde{\pi}(x) \exp\left\{-\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x\, \frac{1}{2}\tilde{\pi}(x)^2\right\} \tag{1.38}$$

So we can directly see it is

$$\langle\phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \mathcal{C} \int_{\phi(0,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x) \exp\left\{\mathrm{i}\int \mathrm{d}^4x\, \mathcal{L}\right\} \tag{1.39}$$

For some reason that we will consider later, the normalization constant $\mathcal{C}$ is ignored in the next calculation.

2.1 Correlation Function

Now we consider the two-point correlation function. For that, we firstly consider the structure

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x)\, \phi(x_1)\phi(x_2) \exp\left\{\mathrm{i}\int_{-T}^{T} \mathrm{d}^4x\, \mathcal{L}\right\}, \tag{1.40}$$

where we assume $x_2^0 > x_1^0$. Now the boundary condition changes from $-T$ to T. So we can separate the integration in three area $-T \to x_1^0$, $x_1^0 \to x_2^0$ and $x_2^0 \to T$ such that we can fix the field at x_1^0 and x_2^0 as $\phi_1(\vec{x})$ and $\phi_2(\vec{x})$ respectively. This operation means no matter how the field evolve from $-T$ to T, we can always separate the integration with 2 steps: First, fixing the field at these two time point and integrate the three area. Second, we integrate all possible situation of these two field.

$$\int \mathcal{D}\phi(x) = \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \int\limits_{\substack{\phi(x_1^0,\vec{x})=\phi_1(\vec{x}) \\ \phi(x_2^0,\vec{x})=\phi_2(\vec{x})}} \mathcal{D}\phi(x). \tag{1.41}$$

So we have

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x)\, \phi(x_1)\phi(x_2) \exp\left\{\mathrm{i}\int_{-T}^{T} \mathrm{d}^4x\, \mathcal{L}\right\}$$
$$= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \int \mathcal{D}\phi(x)\, \phi(x_1)\phi(x_2) \exp\left\{\mathrm{i}\int_{-T}^{T} \mathrm{d}^4x\, \mathcal{L}\right\}. \tag{1.42}$$

Because we have fixed the time point x_1^0 and x_2^0, so $\phi(x_1)$ and $\phi(x_2)$ can be directly picked from the integration $\int \mathcal{D}\phi$ and becomes $\phi_1(\vec{x_1})$ and $\phi_2(\vec{x_2})$, i.e.,

$$\begin{aligned}\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} &\mathcal{D}\phi(x)\,\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\}\\ &=\int\mathcal{D}\phi_1(\vec{x})\int\mathcal{D}\phi_2(\vec{x})\,\phi_1(\vec{x}_1)\phi_2(\vec{x}_2)\left[\int\mathcal{D}\phi(x)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\}\right] &(1.43)\\ &=\int\mathcal{D}\phi_1(\vec{x})\int\mathcal{D}\phi_2(\vec{x})\,\phi_1(\vec{x}_1)\phi_2(\vec{x}_2)\left[\langle\phi_b|e^{-\mathrm{i}\hat{H}(2T)}|\phi_a\rangle\right] &(1.44)\\ &=\int\mathcal{D}\phi_1(\vec{x})\int\mathcal{D}\phi_2(\vec{x})\,\phi_1(\vec{x}_1)\phi_2(\vec{x}_2)\langle\phi_b|e^{-\mathrm{i}\hat{H}(T-x_2^0)}|\phi_2\rangle\langle\phi_2|e^{-\mathrm{i}\hat{H}(x_2^0-x_1^0)}|\phi_1\rangle\langle\phi_1|e^{-\mathrm{i}\hat{H}(x_1^0+T)}|\phi_a\rangle. &(1.45)\end{aligned}$$

Using $\hat{\phi}_S(\vec{x})|\phi_i\rangle=\phi_i(\vec{x})|\phi_i\rangle$ and $\int\mathcal{D}\phi_i|\phi_i\rangle\langle\phi_i|=\mathbb{1}$:

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b}\mathcal{D}\phi(x)\,\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\}=\langle\phi_b|e^{-\mathrm{i}\hat{H}(T-x_2^0)}\hat{\phi}_S(\vec{x}_2)e^{-\mathrm{i}\hat{H}(x_2^0-x_1^0)}\hat{\phi}_S(\vec{x}_1)e^{-\mathrm{i}\hat{H}(x_1^0+T)}|\phi_a\rangle. \tag{1.46}$$

Defining the Heisenberg picture operator $\hat{\phi}_H(x)=e^{\mathrm{i}\hat{H}x^0}\hat{\phi}_S(\vec{x})e^{-\mathrm{i}\hat{H}x^0}$:

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b}\mathcal{D}\phi(x)\,\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\}=\langle\phi_b|e^{-\mathrm{i}\hat{H}T}\hat{\phi}_H(x_2)\hat{\phi}_H(x_1)e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle. \tag{1.47}$$

For a general time ordering $x_2^0>x_1^0$, this becomes the time-ordered product:

$$\langle\phi_b|e^{-\mathrm{i}\hat{H}T}T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle. \tag{1.48}$$

Using the ground state projection

$$\lim_{T\to\infty(1-i\epsilon)}e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle=\lim_{T\to\infty(1-i\epsilon)}\sum_n e^{-\mathrm{i}E_nT}\,|n\rangle\,\langle n|\phi_a\rangle=\langle\Omega|\phi_a\rangle e^{-\mathrm{i}E_0\infty(1-i\epsilon)}|\Omega\rangle, \tag{1.49}$$

and similarly,

$$\lim_{T\to\infty(1-i\epsilon)}\langle\phi_b|\,e^{-\mathrm{i}\hat{H}T}=\lim_{T\to\infty(1-i\epsilon)}\sum_n\langle\phi_b|n\rangle\,\langle n|\,e^{-\mathrm{i}E_nT}=\langle\phi_b|\Omega\rangle\,\langle\Omega|\,e^{-\mathrm{i}E_0\infty(1-i\epsilon)}. \tag{1.50}$$

So we have

$$\langle\phi_b|e^{-\mathrm{i}\hat{H}T}T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle=\langle\phi_b|\phi_a\rangle\langle\Omega|T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}|\Omega\rangle. \tag{1.51}$$

Thus, we can finally reach the correlation function:

$$\langle\Omega|T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}|\Omega\rangle=\frac{\langle\phi_b|e^{-\mathrm{i}\hat{H}T}T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle}{\langle\phi_b|\phi_a\rangle}. \tag{1.52}$$

So in the end, we have

$$\langle\Omega|T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}|\Omega\rangle=\lim_{T\to\infty(1-\mathrm{i}\epsilon)}\frac{\int\mathcal{D}\phi\,\phi(x_1)\phi(x_2)e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}}}{\int\mathcal{D}\phi\,e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}}}. \tag{1.53}$$

2.2 Feynman Rules

Now we want to get the Feynman rules from (1.53), so we need to calculate the numerator and the denominator respectively. For that, we firstly consider the free action of the scalar field:

$$S_0 = \int \mathrm{d}^4x\, \mathcal{L}_0 = \int \mathrm{d}^4x \left(\frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 \right). \tag{1.54}$$

The general Gaussian integration formulas could be useful in the next:

$$\int \mathrm{d}^n x\; e^{-x_i A_{ij} x_j} = \text{const.} \cdot (\det A)^{-1/2}; \tag{1.55}$$

$$\int \mathcal{D}\phi\, e^{\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\, \phi(\partial^2 - m^2)\phi} = \text{const.} \cdot \left(\det\left(\partial^2 - m^2\right)\right)^{-1/2}. \tag{1.56}$$

The idea is firstly discretizing the path integral and then take the limit to get it back to the continuous form. So we work in the lattice $\{x_i\}$, where we can define the discrete field to replace the continuous variable:

$$\mathcal{D}\phi = \prod_i \mathrm{d}\phi\,(x_i). \tag{1.57}$$

And the field can be expressed by the Fourier series:

$$\phi(x_i) = \frac{1}{V}\sum_n e^{-\mathrm{i}k_n \cdot x_i}\phi(k_n); \quad k_n^\mu = \frac{2\pi n^\mu}{L}, \tag{1.58}$$

where n^μ is integers and $|k^\mu| < \pi/\varepsilon$, ε is the distance between two close lattice points. $V = L^4$ is the volume of the space. Because of $\phi(x_n)$ is real, so

$$\begin{aligned} \phi^*(x_i) &= \frac{1}{V}\sum_n e^{+\mathrm{i}k_n^* \cdot x_i}\phi^*(k_n) \\ &= \frac{1}{V}\sum_n e^{-\mathrm{i}(-k_n)\cdot x_i}\phi(-k_n), \end{aligned} \tag{1.59}$$

which means $\phi^*(k_n) = \phi(-k_n)$. So the integral measurement becomes

$$\mathcal{D}\phi(x) = \prod_i \mathrm{d}\phi\,(x_i) = |\det(M)| \prod_n \mathrm{d}\phi\,(k_n) = \prod_{k_n^0 > 0} \mathrm{d}\,\mathrm{Re}\,\phi(k_n)\,\mathrm{d}\,\mathrm{Im}\,\phi(k_n), \tag{1.60}$$

where M is the Fourier transformation matrix, so the Jacobian determinant is 1. And the last equality is because $\mathrm{d}\phi\,(k_n)$ is an area measurement in the complex plane, so this small piece of area measurement is $\mathrm{d}\,\mathrm{Re}\,\phi(k_n)\,\mathrm{d}\,\mathrm{Im}\,\phi(k_n)$. And the free action in momentum space can be rewritten as

$$S_0 = -\frac{1}{V}\sum_{k_n^0 > 0}\frac{1}{2}(m^2 - k_n^2)|\phi_n|^2 = -\frac{1}{V}\sum_{k_n^0 > 0}(m^2 - k_n^2)\left[(\mathrm{Re}\ \phi_n)^2 + (\mathrm{Im}\ \phi_n)^2\right]. \tag{1.61}$$

Therefore, we can together all these parts to express the denominator of (1.53), then using the Gaussian integration formula to calculate the integral:

$$\int \mathcal{D}\phi e^{\mathrm{i}S_0} = \prod_{k_n^0>0} \left(\int \mathrm{d}\,\mathrm{Re}\,\phi_n \exp\left[-\frac{\mathrm{i}}{V}(m^2-k_n^2)(\mathrm{Re}\,\phi_n)^2\right]\right)\left(\int \mathrm{d}\,\mathrm{Im}\,\phi_n \exp\left[-\frac{\mathrm{i}}{V}(m^2-k_n^2)(\mathrm{Im}\,\phi_n)^2\right]\right) \tag{1.62}$$

$$= \prod_{k_n^0>0} \sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}\sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}} \tag{1.63}$$

$$= \prod_{k_n^0>0} \sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}\sqrt{\frac{-\mathrm{i}\pi V}{m^2-(-k_n)^2}} \tag{1.64}$$

$$= \prod_{\text{all } k_n} \sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}, \tag{1.65}$$

where the last step is counting from the upper half plane to the whole complex k_n plane. And to avoid the divergence, we take $k^0 \to k^0(1+\mathrm{i}\epsilon)$ in the denominator. For the expression for the propagator, we still need to calculating $\phi(x_1)\phi(x_2)$ for the numerator of (1.53), so we have

$$\phi(x_1)\phi(x_2) = \frac{1}{V^2}\sum_m e^{-\mathrm{i}k_m x_1}\phi_m \sum_l e^{-\mathrm{i}k_l x_2}\phi_l. \tag{1.66}$$

So the numerator becomes

$$\begin{aligned}\int \mathcal{D}\phi\, \phi(x_1)\phi(x_2) e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}} = \frac{1}{V^2}\sum_{m,l} e^{-\mathrm{i}(k_m x_1 + k_l x_2)} &\left(\prod_{n|k_n^0>0}\int \mathrm{d}\,\mathrm{Re}\,\phi_n\, \mathrm{d}\,\mathrm{Im}\,\phi_n\right) \\ &\times (\mathrm{Re}\,\phi_m + \mathrm{i}\,\mathrm{Im}\,\phi_m)(\mathrm{Re}\,\phi_l + \mathrm{i}\,\mathrm{Im}\,\phi_l) \\ &\times \exp\left\{-\frac{\mathrm{i}}{V}\sum_{n|k_n^0>0}(m^2-k_n^2)\left[(\mathrm{Re}\,\phi_n)^2 + (\mathrm{Im}\,\phi_n)^2\right]\right\}.\end{aligned} \tag{1.67}$$

Notice that for $m \neq l$ and $m \neq -l$, the term

$$(\mathrm{Re}\,\phi_m + \mathrm{i}\,\mathrm{Im}\,\phi_m)(\mathrm{Re}\,\phi_l + \mathrm{i}\,\mathrm{Im}\,\phi_l)\exp\left\{(\mathrm{Re}\,\phi_n)^2 + (\mathrm{Im}\,\phi_n)^2\right\} \tag{1.68}$$

is (**odd function**)$\times$(**even function**) $=$ (**odd function**), so the integral is 0. And for $m = -l$, because of $\phi(-k) = \phi^*(k)$, which means $\phi_l = \phi_{-m} = \phi_m^*$, the production term is then

$$(\phi_m)\cdot(\phi_l) = \phi_m \cdot \phi_m^* = |\phi_m|^2 = (\mathrm{Re}\,\phi_m)^2 + (\mathrm{Im}\,\phi_m)^2. \tag{1.69}$$

So the integration is $\sim \int x^2 e^{-ax^2}$, which is non-zero. Thus only the terms with $k_m = -k_l$ contribute to the integral. Finally, we apply the Gaussian integration formula to (1.67), then we have

$$\int \mathcal{D}\phi\, \phi(x_1)\phi(x_2) e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}} = \frac{1}{V^2}\sum_m e^{-\mathrm{i}k_m(x_1-x_2)}\left(\prod_{k_n^0>0}\frac{-\mathrm{i}\pi V}{m^2-k_n^2}\right)\frac{-\mathrm{i}V}{m^2-k_m^2-\mathrm{i}\varepsilon} \tag{1.70}$$

$$= \frac{1}{V^2}\sum_m e^{-\mathrm{i}k_m(x_1-x_2)}\left(\prod_{\text{all } k_n}\sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}\right)\frac{-\mathrm{i}V}{m^2-k_m^2-\mathrm{i}\varepsilon} \tag{1.71}$$

Finally, we got the Feynman propagator through dividing (1.71) by (1.65):

$$\langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle = \frac{\int \mathcal{D}\phi\, \phi(x_1)\phi(x_2) e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}}}{\int \mathcal{D}\phi\, e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}}} \tag{1.72}$$

$$= \frac{\frac{1}{V^2}\sum_m e^{-\mathrm{i}k_m(x_1-x_2)}\left(\prod_{\text{all } k_n}\sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}\right)\frac{-\mathrm{i}V}{m^2-k_m^2-\mathrm{i}\varepsilon}}{\prod_{\text{all } k_n}\sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}} \tag{1.73}$$

$$= \sum_m \frac{1}{V}\frac{\mathrm{i}}{k_m^2 - m^2 - \mathrm{i}\varepsilon} e^{-\mathrm{i}k_m(x_1-x_2)} \tag{1.74}$$

$$= \int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\varepsilon} e^{-\mathrm{i}k(x_1-x_2)} = D_F(x_1 - x_2) \tag{1.75}$$

2.3 Generating Functional

Now, we consider a more convenient and general method to calculate the Feynman rules. But it might be not very direct to understanding. First, we introduce $J(x)$ as a source in the outside to define the generating functional. For this, it's convenient to review some properties of the functional derivative:

$$\frac{\delta}{\delta J(x)} J(y) = \delta^{(4)}(x-y) \implies \frac{\delta}{\delta J(x)}\int \mathrm{d}^4y\, J(y)\phi(y) = \phi(x). \tag{1.76}$$

And some basic examples of the functional derivative:

1. Exponential function:

$$\frac{\delta}{\delta J(x)} \exp\left\{\mathrm{i}\int \mathrm{d}^4y\, J(y)\phi(y)\right\} = \mathrm{i}\phi(x)\exp\left\{\mathrm{i}\int \mathrm{d}^4y\, J(y)\phi(y)\right\}. \tag{1.77}$$

2. Source terms containing derivatives (obtained through integration by parts):

$$\frac{\delta}{\delta J(x)}\int \mathrm{d}^4y\, \partial_\mu J(y) V^\mu(y) = -\frac{\delta}{\delta J(x)}\int \mathrm{d}^4y\, J(y)\partial_\mu V^\mu(y) = -\partial_\mu V^\mu(x). \tag{1.78}$$

3. Functional composite differentiation (chain rule):

$$\frac{\delta F[\phi]}{\delta J(x)} = \int \mathrm{d}^4z\, \frac{\delta F[\phi]}{\delta\phi(z)}\frac{\delta\phi(z)}{\delta J(x)} \quad \left(\text{Generalization of } \frac{\partial x_i(z_k)}{\partial y_j} = \sum_k \frac{\partial x_i}{\partial z_k}\frac{\partial z_k}{\partial y_j}\right). \tag{1.79}$$

Define the **Generating Functional** $Z[J]$ as:

$$Z[J] = \int \mathcal{D}\phi \exp\left\{\mathrm{i}\int \mathrm{d}^4x\, (\mathcal{L} + J(x)\phi(x))\right\}. \tag{1.80}$$

Using $Z[J]$, we can conveniently calculate the n-point correlation function:

$$\langle\Omega|\, T\{\phi(x_1)\phi(x_2)\}\,|\Omega\rangle = \frac{1}{Z[0]}\left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right) Z[J]\Big|_{J=0} \tag{1.81}$$

2.4 Feynman Rules for the Free Theory

For a free scalar field, the Lagrangian is

$$\mathcal{L}_0 = \frac{1}{2}\left[(\partial_\mu\phi)(\partial^\mu\phi) + (-m^2 + \mathrm{i}\epsilon)\phi^2\right] \tag{1.82}$$
$$= \frac{1}{2}\left[\partial_\mu(\phi\,\partial^\mu\phi) - (\phi\,\partial^2\phi) + (-m^2 + \mathrm{i}\epsilon)\phi^2\right] \tag{1.83}$$
$$= \frac{1}{2}\phi(-\partial^2 - m^2 + \mathrm{i}\epsilon)\phi. \tag{1.84}$$

Define the operator $K_x = -\partial^2 - m^2 + \mathrm{i}\epsilon$, and the action with source can be written as:

$$S_J = \int \mathrm{d}^4x\,(\mathcal{L}_0 + J\phi) = \int \mathrm{d}^4x \left(\frac{1}{2}\phi K_x\phi + J\phi\right) \tag{1.85}$$
$$= \int \mathrm{d}^4x\,\mathrm{d}^4y \left(\frac{1}{2}\phi(x)K(x,y)\phi(y)\right) + \int \mathrm{d}^4x\,J(x)\phi(x), \tag{1.86}$$

where $K(x,y) = \delta^{(4)}(x-y)K_y$. Now the generating functional is:

$$Z_0[J] = \int \mathcal{D}\phi \exp\left\{\mathrm{i}\int \mathrm{d}^4x \left(\frac{1}{2}\phi K\phi + J\phi\right)\right\}. \tag{1.87}$$

Now we change the variable $\phi \to \phi' = \phi + K^{-1}J$, so the measurement keeps unchanged $\mathcal{D}\phi' = \mathcal{D}\phi$:

$$Z_0[J] = \int \mathcal{D}\phi' \left|\det\frac{\delta\phi'}{\delta\phi}\right| \exp\left\{\mathrm{i}\int \mathrm{d}^4x \left[\frac{1}{2}(\phi' - K^{-1}J)K(\phi' - K^{-1}J) + J(\phi' - K^{-1}J)\right]\right\} \tag{1.88}$$
$$= \left(\int \mathcal{D}\phi' \exp\left\{\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\phi' K\phi'\right\}\right) \exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\,J(x)K^{-1}(x,y)J(y)\right\} \tag{1.89}$$
$$= Z_0[J=0] \exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\,J(x)K^{-1}(x,y)J(y)\right\}, \tag{1.90}$$

where $\left|\det\frac{\delta\phi'}{\delta\phi}\right| = 1$. And notice for

$$J(y) = K(x,y)\left(K^{-1}(x,y)J(y)\right) \tag{1.91}$$
$$= \left(-\partial_x^2 - m^2 + \mathrm{i}\epsilon\right) K^{-1}(x,y)\delta^{(4)}(x-y)J(y) \tag{1.92}$$
$$= \left(-\partial_x^2 - m^2 + \mathrm{i}\epsilon\right) K^{-1}(x,y)J(x). \tag{1.93}$$

Thus, K^{-1} satisfies the Green's function (except an $-\mathrm{i}$ factor in the right hand side):

$$(-\partial_x^2 - m^2 + \mathrm{i}\epsilon)K^{-1}(x,y) = \delta^{(4)}(x-y). \tag{1.94}$$

Take a sharp look! So we can directly realize that the Feynman propagator $-\mathrm{i}D_F(x-y)$ also satisfies this equation, so we realize that $K^{-1}(x-y)$ is nothing but $-\mathrm{i}D_F(x-y)$. So the **generating functional in free theory** can be then expressed as

$$Z_0[J] = Z_0[0] \exp\left\{-\frac{1}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\,J(x)D_F(x-y)J(y)\right\}. \tag{1.95}$$

So the correlation function can be obtained from (1.81):

$$\langle 0|\, T\{\phi(x_1)\phi(x_2)\}\, |0\rangle = \frac{1}{Z_0[0]}\left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right) Z_0[J]\Big|_{J=0} \tag{1.96}$$

$$= \left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right)\exp\left\{-\frac{1}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\, J(x)D_F(x-y)J(y)\right\}\Big|_{J=0} \tag{1.97}$$

$$= \left(-\frac{\delta}{\delta J(x_1)}\right)\left[-\frac{1}{2}\left(\int \mathrm{d}^4y\, D_F(x_2-y)J(y) + \int \mathrm{d}^4x\, J(x)D_F(x-x_2)\right)\right] \times \exp\left\{-\frac{1}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\, J(x)D_F(x-y)J(y)\right\}\Big|_{J=0} \tag{1.98}$$

$$= \frac{\delta}{\delta J(x_1)}\left[\int \mathrm{d}^4y\, D_F(y-x_2)J(y)\right]\frac{Z_0[J]}{Z_0[0]}\Big|_{J=0} \tag{1.99}$$

$$= D_F(x_1-x_2)\,\frac{Z_0[J]}{Z_0[0]}\Big|_{J=0} + \left[\int \mathrm{d}^4y\, D_F(y-x_2)J(y)\right]\left[\int \mathrm{d}^4y\, D_F(y-x_1)J(y)\right]\frac{Z_0[J]}{Z_0[0]}\Big|_{J=0} \tag{1.100}$$

$$= D_F(x_1-x_2). \tag{1.101}$$

Technically, the last step we do, which impose the boundary condition to cancel the second term is not very strict. In general, one can not impose the boundary condition before actually calculating the integration. More precisely, the reason why we have confidence to do so is based on the property that we can let $f \in L^2(\mathbb{R}^{1,3})$ be fixed, and define a functional

$$F_f : L^2(\mathbb{R}^{1,3}) \to \mathbb{C}, \qquad F_f[J] := \int_{\mathbb{R}^{1,3}} f(x)\, J(x)\, dx. \tag{1.102}$$

Then F_f is a continuous linear functional on $L^2(\mathbb{R}^{1,3})$. Indeed, by the Cauchy–Schwarz inequality,

$$|F_f[J]| \le \|f\|_{L^2}\, \|J\|_{L^2}, \tag{1.103}$$

so F_f is bounded and hence continuous. In particular, for any sequence (or net) $J_n \to 0$ in $L^2(\mathbb{R}^{1,3})$, one has

$$\lim_{n\to\infty} F_f[J_n] = F_f[0] = 0. \tag{1.104}$$

Therefore, the notation$F_f[J]\,|_{J=0}$ is to be understood as the evaluation of the functional F_f at the zero element of $L^2(\mathbb{R}^{1,3})$, i.e.

$$F_f[J]\,|_{J=0} := F_f(0). \tag{1.105}$$

Moreover, by continuity, this evaluation coincides with the limit along any sequence $J_n \to 0$ in the L^2 topology:

$$F_f(0) = \lim_{J\to 0} F_f[J], \tag{1.106}$$

where the limit is taken with respect to the L^2 norm.

2.5 Interaction theory: ϕ^4 theory as an example

For theories involving interactions, such as $\mathcal{L} = \mathcal{L}_0 - \frac{\lambda}{4!}\phi^4$ and $\mathcal{L}_0 = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2$, the generating functional can be written as:

$$Z[J] = \int \mathcal{D}\phi \exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4 + \mathcal{L}_0 + J(x)\phi(x)\right)\right\}. \tag{1.107}$$

Notice that

$$\begin{aligned}\frac{\delta}{\delta J(x)}Z_0[J] &= \frac{\delta}{\delta J(x)}\int \mathcal{D}\phi \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\,\mathcal{L}_0 + J(x)\phi(x)\right\} \\ &= \phi(x)\int \mathcal{D}\phi \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\,\mathcal{L}_0 + J(x)\phi(x)\right\} \\ &= \phi(x)Z_0[J]. \end{aligned} \tag{1.108}$$

So we have

$$\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4 Z_0[J] = \phi^4 Z_0[J], \tag{1.109}$$

which means

$$Z[J] = \int \mathcal{D}\phi \left\{1 + \left(-\mathrm{i}\int \mathrm{d}^4x\,\frac{\lambda}{4!}\phi^4\right) + \frac{1}{2!}\left(-\mathrm{i}\int \mathrm{d}^4x\,\frac{\lambda}{4!}\phi^4\right)^2 + \dots\right\}\exp\left\{\mathrm{i}\int \mathrm{d}^4x\,[\mathcal{L}_0 + J(x)\phi(x)]\right\} \tag{1.110}$$

$$= \int \mathcal{D}\phi \left\{1 + \left[-\mathrm{i}\int \mathrm{d}^4x\,\frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right] + \frac{1}{2!}\left[-\mathrm{i}\int \mathrm{d}^4x\,\frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right]^2 + \dots\right\} \times \exp\left\{\mathrm{i}\int \mathrm{d}^4x\,[\mathcal{L}_0 + J(x)\phi(x)]\right\} \tag{1.111}$$

$$= \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\,\frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right\}\int \mathcal{D}\phi \exp\left\{\mathrm{i}\int \mathrm{d}^4x\,[\mathcal{L}_0 + J(x)\phi(x)]\right\} \tag{1.112}$$

$$= \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\,\frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right\} Z_0[J], \tag{1.113}$$

where we expanded the exponential term of interaction part as the perturbed series, and using the derivative relation (1.109). So we link the interaction theory generating functional with the free theory generating functional:

$$Z[J] = \left(1 - \frac{\mathrm{i}\lambda}{4!}\int \mathrm{d}^4x\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4 + \dots\right) Z_0[J]. \tag{1.114}$$

So the correlation function can be expanded by the free propagator as

$$\langle\Omega|T\{\phi(x_1)\phi(x_2)\}|\Omega\rangle = \frac{1}{Z[0]}\left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right)Z[J]\Big|_{J=0} \tag{1.115}$$

$$= \frac{1}{Z[0]}\left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right)\left(1 - \frac{\mathrm{i}\lambda}{4!}\int \mathrm{d}^4x\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4 + \dots\right) Z_0[J]\Big|_{J=0} \tag{1.116}$$

$$= \frac{1}{Z[0]}\left[(-\mathrm{i})^2\frac{\delta^2 Z_0[J]}{\delta J(x_1)\delta J(x_2)} + \left(-\frac{\mathrm{i}\lambda}{4!}\right)\int \mathrm{d}^4x\,(-\mathrm{i})^6\,\frac{\delta^6 Z_0[J]}{\delta J(x)^4\delta J(x_1)\delta J(x_2)} + \dots\right]\Bigg|_{J=0} \tag{1.117}$$

$$= \frac{1}{Z[0]}\langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle + \frac{1}{Z[0]}\left(-\mathrm{i}\frac{\lambda}{4!}\right)\int \mathrm{d}^4x\,\langle 0|T\{\phi^4(x)\phi(x_1)\phi(x_2)\}|0\rangle + \dots \tag{1.118}$$

$$= \frac{1}{Z[0]}\langle 0|T\left\{\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4\right)\right\}\right\}|0\rangle. \tag{1.119}$$

Comparing with the result we got in the canonical quantization

$$\langle\Omega| T\{\phi(x_1)\phi(x_2)\} |\Omega\rangle = \frac{\langle 0| T\left\{\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4\right)\right\}\right\}|0\rangle}{\langle 0| T\left\{\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4\right)\right\}\right\}|0\rangle}, \tag{1.120}$$

we find in the interaction case, $Z[0]$ has different meaning comparing with $Z_0[0]$ in the free case, i.e.,

$$Z[0] = T\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4\right)\right\}, \tag{1.121}$$

which included all the **Vacuum Bubbles**.

2.6 Other Type of Generating functional

We define the **Energy Functional** for the connected graphs $W[J]$ as

$$W[J] = \mathrm{i}\ln Z[J]. \tag{1.122}$$

So we have

$$\frac{\delta W[J]}{\delta J(x)} = \frac{\mathrm{i}}{Z[0]}\frac{\delta Z[J]}{\delta J(x)} = -\frac{\langle\Omega|\phi(x)|\Omega\rangle}{\langle\Omega|\Omega\rangle} = -\phi_{\text{classical}}(x), \tag{1.123}$$

where the second step is because $\delta Z[J] = \langle\Omega|\mathrm{i}\phi(x)|\Omega\rangle$. And in physics, it is defined as the **Classical Field**. And the definition of $W[J]$ can be easily seen that it's very similar with the free energy in classical statistical mechanics with the partition function $Z[J]$. So using the Legendre transformation, we can define the **Effective Action**, which looks like the Gibbs free energy $\Gamma[\phi_{cl}]$ as

$$\Gamma[\phi_{cl}] = -W[J] - \int \mathrm{d}^4x\, J(x)\phi_{cl}(x). \tag{1.124}$$

Base on this, we can see there is a symmetric relation:

$$\frac{\delta\Gamma[\phi_{cl}]}{\delta\phi_{cl}(x)} = -J(x); \quad \frac{\delta W[J]}{\delta J(x)} = -\phi_{cl}(x). \tag{1.125}$$

Taking another functional derivative with respect to $J(y)$ on both sides yields:

$$\Rightarrow \frac{\delta}{\delta J(y)}\frac{\delta}{\delta\Phi_{\rm cl}(x)}\Gamma[\Phi_{\rm cl}] = -\frac{\delta J(x)}{\delta J(y)} = -\delta(x-y). \tag{1.126}$$

Using the chain rule for functional differentiation $\frac{\delta}{\delta J(y)} = \int \mathrm{d}^4z\,\frac{\delta\Phi_{\rm cl}(z)}{\delta J(y)}\frac{\delta}{\delta\Phi_{\rm cl}(z)}$, we obtain the following identity:

$$-\delta(x-y) = \int \mathrm{d}^4z\,\frac{\delta\Phi_{\rm cl}(z)}{\delta J(y)}\frac{\delta^2\Gamma[\Phi_{\rm cl}]}{\delta\Phi_{\rm cl}(z)\delta\Phi_{\rm cl}(x)}. \tag{1.127}$$

We can rewrite the first term as $\frac{\delta\Phi_{\rm cl}(z)}{\delta J(y)} = \frac{\delta^2 W[J]}{\delta J(y)\delta J(z)}$, where $W[J]$ is the generating functional of connected Green's functions:

$$-\delta(x-y) = \int \mathrm{d}^4z\,\underbrace{\frac{\delta^2 W[J]}{\delta J(y)\delta J(z)}}_{-\mathrm{i}G(y-z)}\underbrace{\frac{\delta^2\Gamma[\Phi_{\rm cl}]}{\delta\Phi_{\rm cl}(z)\delta\Phi_{\rm cl}(x)}}_{\mathrm{i}G^{-1}(z-x)}. \tag{1.128}$$

Here, the first part corresponds to the connected two-point function $-\mathrm{i}G(y-z)$, and the second part corresponds to the inverse propagator $\mathrm{i}G^{-1}(z-x)$. Thus, the equation simplifies to:

$$\delta(x-y) = \int \mathrm{d}^4z \, [-\mathrm{i}G(y-z)] \left[\mathrm{i}G^{-1}(z-x)\right]. \tag{1.129}$$

The n-th derivative of $W[J]$ gives the connected correlation function

$$\left.\frac{\delta^n W[J]}{\delta J(x_1)\dots\delta J(x_n)}\right|_{J=0} = (\mathrm{i})^{n+1} \langle\Omega| \, T\{\phi(x_1)\dots\phi(x_n)\} \, |\Omega\rangle_{connect}. \tag{1.130}$$

And the n-th derivative of $\Gamma[\phi_{cl}]$ gives the correlation function for 1PI graphs

$$\frac{\delta^n \Gamma[\phi_{cl}]}{\delta\phi_{cl}(x_1)\dots\delta\phi_{cl}(x_n)} = -\mathrm{i} \, \langle\Omega| \, T\{\phi(x_1)\dots\phi(x_n)\} \, |\Omega\rangle_{1PI}. \tag{1.131}$$

3. Quantization of EM Field in Path Integral Formalism

The classical action for the electromagnetic field is given by:

$$S[A] = -\frac{1}{4}\int \mathrm{d}^4x \, F_{\mu\nu}F^{\mu\nu}, \tag{1.132}$$

where $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. Integrating by parts, we can rewrite the action as:

$$S[A] = \frac{1}{2}\int \mathrm{d}^4x \, A_\mu D^{\mu\nu} A_\nu, \quad \text{with} \quad D^{\mu\nu} = g^{\mu\nu}\Box - \partial^\mu\partial^\nu. \tag{1.133}$$

In the path integral formulation, the vacuum-to-vacuum transition amplitude is given by:

$$\int \mathcal{D}A \, e^{\mathrm{i}S[A]}. \tag{1.134}$$

But the problem the measure $\mathcal{D}A = \mathcal{D}A^0\mathcal{D}A^1\mathcal{D}A^2\mathcal{D}A^3$ is **not** well defined. The integral is ill-defined because of the redundancy due to infinite gauge copies. Under a local gauge transformation:

$$A_\mu \to A_\mu^\alpha = A_\mu + \frac{1}{e}\partial_\mu\alpha(x), \tag{1.135}$$

where $\alpha(x)$ is an arbitrary scalar function. For a pure gauge choice where $A_\mu = \frac{1}{e}\partial_\mu\alpha$, we find that the field strength $F_{\mu\nu} = 0$, leading to:

$$S[A] = 0 \quad \Rightarrow \quad e^{\mathrm{i}S[A]} = 1. \tag{1.136}$$

This implies that we have an unbounded path integral because the integrand does not decay along the gauge orbits.

3.1 Faddeev-Popov Quantization

Solution: We need to divide out the gauge degrees of freedom in the measure $\mathcal{D}A$. This procedure is known as the **Faddeev-Popov quantization**.

Let $G(A)$ be a function that fixes the gauge (for example, $G(A) = \partial_\mu A^\mu = 0$ for the Lorentz gauge). We want to define $\int \mathcal{D}A \, e^{\mathrm{i}S[A]}$ with a constraint such that the path integral only sums over field configurations fulfilling $G(A) = 0$.

To motivate this, recall the standard Gauss integration with matrix M (rank N):

$$(\det M)^{-1/2} = (2\pi)^{-N/2} \int_{-\infty}^{+\infty} \mathrm{d}x_1 \dots \mathrm{d}x_N \, e^{-\frac{1}{2}x^T M x}. \tag{1.137}$$

Generalizing this to a functional determinant (Path integral):

$$(\det M)^{-1/2} = \int \mathcal{D}\phi \, e^{-\frac{1}{2}\int \mathrm{d}^4x \phi(x) M(x) \phi(x)}, \tag{1.138}$$

where M is some differential operator, e.g., $M(x) = \Box + m_\phi^2$.

Next, consider the Dirac δ-distribution on an N-dimensional space for an arbitrary function $\vec{g}(x_1, \dots, x_N)$:

$$\int_{-\infty}^{+\infty} \mathrm{d}x_1 \dots \mathrm{d}x_N \, \delta^{(N)} \left[\vec{g}(x_1, \dots, x_N)\right] \det \left[\frac{\partial g_i}{\partial x_j}\right] = 1 \tag{1.139}$$

In 1-dimension, this reduces to the property:

$$\int \mathrm{d}x \, \delta(ax) a = \int \mathrm{d}x \, \frac{1}{a}\delta(x) a = 1. \tag{1.140}$$

Extending this to a functional Dirac distribution for gauge fields, we have the Faddeev-Popov identity:

$$\int \mathcal{D}\alpha(x) \underbrace{\delta\left[G(A^\alpha)\right] \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right]}_{\text{Faddeev-Popov det.}} = 1. \tag{1.141}$$

Quantization of electrodynamics proceeds by inserting this identity "1" into the path integral:

$$I = \int \mathcal{D}A \, e^{\mathrm{i}S[A]} = \int \mathcal{D}A (1) e^{\mathrm{i}S[A]} = \int \mathcal{D}A \left(\int \mathcal{D}\alpha \, \delta\left[G(A^\alpha)\right] \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \right) e^{\mathrm{i}S[A]}. \tag{1.142}$$

If $G(A^\alpha)$ is the Lorentz gauge condition, we have:

$$G(A^\alpha) = \partial_\mu A^\mu + \frac{1}{e}\Box\alpha. \tag{1.143}$$

Then the functional derivative becomes:

$$\det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] = \det \left(\frac{\Box}{e}\right). \tag{1.144}$$

Notice that in quantum electrodynamics, this determinant is invariant under gauge transformations and independent of A_μ.

Due to gauge invariance, we have $\mathcal{D}A = \mathcal{D}A^\alpha$ and $S[A] = S[A^\alpha]$. Therefore, the path integral can be rewritten as:

$$\begin{aligned} I &= \int \mathcal{D}\alpha \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \int \mathcal{D}A \, \delta\left[G(A^\alpha)\right] e^{\mathrm{i}S[A]} \\ &= \int \mathcal{D}\alpha \left(\det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \int \mathcal{D}A^\alpha \, \delta\left[G(A^\alpha)\right] e^{\mathrm{i}S[A^\alpha]} \right) \\ &= \int \mathcal{D}\alpha \, \tilde{I}. \end{aligned} \tag{1.145}$$

By dropping the infinite volume factor of the gauge group $\int \mathcal{D}\alpha$, we define the physical path integral for QED as:

$$\tilde{I} = \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \int \mathcal{D}A^\alpha \, \delta\left[G(A^\alpha)\right] e^{\mathrm{i}S[A]}. \tag{1.146}$$

3.2 Gauge Fixing and the Photon Propagator

Our next task is to absorb the Dirac constraint $\delta[G(A^\alpha)]$ into a modified action. Let us generalize the gauge fixing condition by setting:

$$G(A^\alpha) = \partial^\mu A_\mu(x) - \omega(x), \tag{1.147}$$

where $\omega(x)$ is an arbitrary scalar function. We also set the determinant remains independent of ω as the form:

$$\det\left[\frac{\delta G(A^\alpha)}{\delta\alpha}\right] = \det\left(\frac{\Box}{e}\right). \tag{1.148}$$

So the functional integration $\tilde{I}$ becomes

$$\int \mathcal{D}A e^{-\mathrm{i}S[A]} = \det\left(\frac{\Box}{e}\right)\int \mathcal{D}A^\alpha\, \delta\left[\partial^\mu A_\mu - \omega(x)\right] e^{\mathrm{i}S[A]}. \tag{1.149}$$

Since this formula is satisfied for any $\omega(x)$, so we can integrating over all possible functions $\omega(x)$ with a Gaussian weight centered at $\omega = 0$:

$$I' = N(\xi)\int \mathcal{D}\omega\, e^{-\mathrm{i}\int \mathrm{d}^4x \frac{\omega(x)^2}{2\xi}}\,\tilde{I}, \tag{1.150}$$

where $N(\xi)$ is a normalization factor. Then we substitute $\tilde{I}$ and explicitly show the delta function:

$$\begin{aligned}
I' &= N(\xi)\det\left(\frac{\Box}{e}\right)\int \mathcal{D}A\, e^{\mathrm{i}S[A]}\int \mathcal{D}\omega\, e^{-\mathrm{i}\int \mathrm{d}^4x\frac{\omega^2}{2\xi}}\quad \underbrace{\delta[\partial^\mu A_\mu - \omega]}_{\text{linear covariant gauge}} \\
&= N(\xi)\det\left(\frac{\Box}{e}\right)\int \mathcal{D}A\, e^{\mathrm{i}S[A]}e^{-\frac{\mathrm{i}}{2\xi}\int \mathrm{d}^4x(\partial^\mu A_\mu)^2} \\
&= N(\xi)\det\left(\frac{\Box}{e}\right)\int \mathcal{D}A\, e^{\mathrm{i}S'[A]}.
\end{aligned} \tag{1.151}$$

The modified action $S'[A]$ becomes:

$$S'[A] = \frac{1}{2}\int \mathrm{d}^4x\left[A_\mu D^{\mu\nu}A_\nu - \frac{1}{\xi}(\partial^\mu A_\mu)^2\right] = \frac{1}{2}\int \mathrm{d}^4x\, A_\mu \tilde{D}^{\mu\nu}A_\nu, \tag{1.152}$$

which implies the new differential operator:

$$\tilde{D}^{\mu\nu} = g^{\mu\nu}\Box - \left(1 - \frac{1}{\xi}\right)\partial^\mu\partial^\nu. \tag{1.153}$$

From this modified action, we can derive the generating functional as

$$Z[J_\mu] = \int \mathcal{D}A \exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(\frac{1}{2}A_\mu\tilde{D}^{\mu\nu}A_\nu + J^\mu(x)A_\mu(x)\right)\right\}. \tag{1.154}$$

Then transform the field as $A_\mu \to A'_\mu = A_\mu + \tilde{D}_\mu^{-1\nu}J_\nu$, we have

$$Z[J_\mu] = \int \mathcal{D}A\, \exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(\frac{1}{2}\left(A'_\mu - \tilde{D}_\mu^{-1\lambda}J_\lambda\right)\tilde{D}^{\mu\nu}\left(A'_\nu - \tilde{D}_\nu^{-1\lambda}J_\lambda\right) + J^\mu(x)\left(A'_\mu - \tilde{D}_\mu^{-1\lambda}J_\lambda\right)\right)\right\} \tag{1.155}$$

$$= \left(\int \mathcal{D}A'\, \exp\left\{\frac{\mathrm{i}}{2}A'_\mu\tilde{D}^{\mu\nu}A'_\nu\right\}\right)\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\left(J_\mu(x)\tilde{D}^{-1\mu\nu}J_\nu(y)\right)\right\} \tag{1.156}$$

$$= Z[0]\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\left(J_\mu(x)\tilde{D}^{-1\mu\nu}J_\nu(y)\right)\right\} \tag{1.157}$$

$$= Z[0]\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\left(J_\mu(x)\tilde{G}^{\mu\nu}J_\nu(y)\right)\right\}, \tag{1.158}$$

where we have ignored the determinant due to $\det\left(\frac{\delta A'}{\delta A}\right) = 1$. And the photon propagator is the inverse of $\tilde{D}^{\mu\nu}$ in momentum space:

$$G^{\mu\nu}(x-y) = -\mathrm{i}\int \frac{\mathrm{d}^4k}{(2\pi)^4} e^{-\mathrm{i}k(x-y)} \frac{g^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{k^2}}{k^2 + \mathrm{i}\epsilon}. \tag{1.159}$$

The effective action then can be constructed as same as (1.124)

$$\Gamma[A^\mu_{cl}] = -\mathrm{i}Z[J_\mu] - \int \mathrm{d}^4x\, J_\mu(x) A^\mu_{cl}. \tag{1.160}$$

$S'[A]$ with $\tilde{D}^{\mu\nu}$ now has no problems with the longitudinal degrees of freedom for photons.

3.3 Standard Gauge Choices

Depending on the choice of the parameter ξ and conditions, we have different gauges in the linear covariant gauge family:

- $\xi = 0$: **Landau gauge**, corresponding to the condition $\partial^\mu A_\mu = 0$.
- $\xi = 1$: **Feynman gauge**.
- $i = 1, 2, 3, \quad \partial^i A_i = 0$: **Coulomb gauge**.
- $n^\mu A_\mu = 0$: **Axial gauge**.

Finally, the path integral in QED for calculating an n-point correlator is written as:

$$\langle\Omega|T\,\hat{\mathcal{O}}(A)|\Omega\rangle = \lim_{T\to\infty(1-\mathrm{i}\epsilon)} \frac{\int \mathcal{D}A\,\hat{\mathcal{O}}(A)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\left[\mathcal{L}(x) - \frac{1}{2\xi}(\partial^\mu A_\mu)^2\right]\right\}}{\int \mathcal{D}A\,\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\left[\mathcal{L}(x) - \frac{1}{2\xi}(\partial^\mu A_\mu)^2\right]\right\}}. \tag{1.161}$$

4. Path Integral for Fermions and Dirac Spinors

For a scalar field the path integral is an ordinary Gaussian integral repeated at every spacetime point. For a fermion field this cannot be right, because the quantum operators obey anti-commutation relations. The path-integral version of this anti-commuting nature is to let the classical integration variables themselves be Grassmann numbers. Thus in the free Dirac theory we formally write

$$Z_0 = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi\,\exp\left\{\mathrm{i}\int \mathrm{d}^4x\,\bar{\psi}(x)\big(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon\big)\psi(x)\right\}. \tag{1.162}$$

Here ψ and $\bar{\psi}$ are treated as independent Grassmann-valued integration variables. The relation $\bar{\psi} = \psi^\dagger\gamma^0$ is the operator relation in canonical quantization; in the functional integral the two variables are integrated independently and the correct conjugation properties are recovered in correlation functions.

4.1 Grassmann algebra

A set of Grassmann numbers θ_i is defined by

$$\theta_i\theta_j = -\theta_j\theta_i, \qquad \theta_i^2 = 0. \tag{1.163}$$

Because every variable squares to zero, a function of one Grassmann variable stops after the linear term:

$$f(\theta) = a + b\theta. \tag{1.164}$$

We use the left derivative convention. The basic rule is

$$\frac{\partial}{\partial\theta}\theta = 1, \qquad \frac{\partial}{\partial\theta}(AB) = \frac{\partial A}{\partial\theta}B + (-1)^{|A|}A\frac{\partial B}{\partial\theta}, \tag{1.165}$$

where $|A| = 0$ if A is Grassmann-even and $|A| = 1$ if A is Grassmann-odd. For example, if b is Grassmann-odd, then

$$\frac{\partial}{\partial\theta}(b\theta) = -b. \tag{1.166}$$

The minus sign is not optional; it is the same sign that will later produce the minus signs of fermion loops and fermion contractions.

Berezin integration is defined so that it is the same operation as differentiation:

$$\int \mathrm{d}\theta\ 1 = 0, \qquad \int \mathrm{d}\theta\ \theta = 1. \tag{1.167}$$

For many variables the order matters. With the convention that the rightmost integration is performed first,

$$\int \mathrm{d}\theta_1\,\mathrm{d}\theta_2\ \theta_1\theta_2 = \int \mathrm{d}\theta_1 \left(\int \mathrm{d}\theta_2\ \theta_1\theta_2\right) = \int \mathrm{d}\theta_1\,(-\theta_1) = -1. \tag{1.168}$$

This simple example is often the safest way to remember that Grassmann integration is sensitive to ordering.

The finite-dimensional Gaussian integral is the key formula. For one pair $\eta, \bar{\eta}$,

$$\int \mathrm{d}\bar{\eta}\,\mathrm{d}\eta\ \exp(-\bar{\eta}b\eta) = \int \mathrm{d}\bar{\eta}\,\mathrm{d}\eta\ (1 - \bar{\eta}b\eta) = \int \mathrm{d}\bar{\eta}\,\mathrm{d}\eta\ (1 + \eta\bar{\eta}b) = b. \tag{1.169}$$

With an insertion one obtains

$$\int \mathrm{d}\bar{\eta}\,\mathrm{d}\eta\ \eta\bar{\eta}\ \exp(-\bar{\eta}b\eta) = 1 = b^{-1}\det b. \tag{1.170}$$

For N pairs this becomes

$$\int \prod_{i=1}^{N} \mathrm{d}\bar{\eta}_i\,\mathrm{d}\eta_i\ \exp\left(-\bar{\eta}_i A_{ij}\eta_j\right) = \det A, \tag{1.171}$$

and the two-point insertion gives the inverse matrix:

$$\int \prod_{i=1}^{N} \mathrm{d}\bar{\eta}_i\,\mathrm{d}\eta_i\ \eta_j\bar{\eta}_k\ \exp\left(-\bar{\eta}_i A_{i\ell}\eta_\ell\right) = (A^{-1})_{jk}\det A. \tag{1.172}$$

This is the fermionic analogue of the ordinary bosonic Gaussian, but the determinant appears in the numerator rather than in the denominator. In parallel,

$$\int \mathrm{d}^N x\ \exp\left(-\frac{1}{2}x_i A_{ij} x_j + J_i x_i\right) = \text{const.}\,(\det A)^{-1/2}\exp\left(\frac{1}{2}J_i (A^{-1})_{ij} J_j\right), \tag{1.173}$$

$$\int \prod_i \mathrm{d}\bar{\eta}_i\,\mathrm{d}\eta_i\ \exp\left(-\bar{\eta}_i A_{ij}\eta_j + \bar{\xi}_i\eta_i + \bar{\eta}_i\xi_i\right) = \det A\ \exp\left(\bar{\xi}_i (A^{-1})_{ij}\xi_j\right). \tag{1.174}$$

The inverse power in the bosonic case and the direct determinant in the fermionic case are not cosmetic differences; this is why closed fermion loops contribute an extra minus sign and a trace over spinor indices.

The change of variables also goes oppositely to the bosonic case. If

$$\theta_i = U_{ij}\theta'_j, \tag{1.175}$$

then the product of all Grassmann variables transforms as

$$\theta_1\theta_2\cdots\theta_N = (\det U)\,\theta'_1\theta'_2\cdots\theta'_N. \tag{1.176}$$

Since the integral of the top product must remain normalized to one, the measure transforms as

$$\mathrm{d}\theta_N\cdots\mathrm{d}\theta_1 = (\det U)^{-1}\,\mathrm{d}\theta'_N\cdots\mathrm{d}\theta'_1\,. \tag{1.177}$$

This inverse Jacobian is the finite-dimensional origin of fermionic functional determinants in field theory.

4.2 Fermion field propagator

Now apply the same Gaussian formula to a Dirac field. A Dirac spinor is a four-component column of Grassmann-valued fields,

$$\psi(x) = \begin{pmatrix}\psi_1(x)\\ \psi_2(x)\\ \psi_3(x)\\ \psi_4(x)\end{pmatrix}, \qquad \bar\psi(x) = \psi^\dagger(x)\gamma^0, \tag{1.178}$$

and the free Dirac Lagrangian is

$$\mathcal{L}_{\text{Dirac}} = \bar\psi_a(x)(\mathrm{i}\gamma^\mu_{ab}\partial_\mu - m\delta_{ab})\psi_b(x). \tag{1.179}$$

After discretizing spacetime, the labels (x,a) play the role of the finite-dimensional index i. Therefore the free vacuum functional is the continuum version of a Grassmann Gaussian:

$$Z_0[0,0] = \int \mathcal{D}\bar\psi\mathcal{D}\psi\,\exp\left[\mathrm{i}\int \mathrm{d}^4x\ \bar\psi(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon)\psi\right] = \text{const.}\,\det\big(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon\big). \tag{1.180}$$

The constant depends on the precise normalization of the measure and will cancel in normalized correlation functions.

To compute correlation functions, introduce Grassmann sources $\eta, \bar\eta$:

$$Z_0[\eta,\bar\eta] = \int \mathcal{D}\bar\psi\mathcal{D}\psi\,\exp\left\{\mathrm{i}\int \mathrm{d}^4x\,\left[\bar\psi(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon)\psi + \bar\psi\eta + \bar\eta\psi\right]\right\}. \tag{1.181}$$

Let

$$D = \mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon. \tag{1.182}$$

Completing the square gives

$$\bar\psi D\psi + \bar\psi\eta + \bar\eta\psi = \big(\bar\psi + \bar\eta D^{-1}\big)D\big(\psi + D^{-1}\eta\big) - \bar\eta D^{-1}\eta. \tag{1.183}$$

The shift of Grassmann integration variables has unit Jacobian, so

$$Z_0[\eta,\bar{\eta}] = Z_0[0,0]\,\exp\left[-\mathrm{i}\int \mathrm{d}^4x\,\mathrm{d}^4y\;\bar{\eta}(x)D^{-1}(x,y)\eta(y)\right]. \tag{1.184}$$

It is conventional to define the Feynman propagator by

$$S_F(x-y) = \mathrm{i}D^{-1}(x,y), \tag{1.185}$$

or equivalently

$$\left(\mathrm{i}\not{\partial}_x - m + \mathrm{i}\epsilon\right)S_F(x-y) = \mathrm{i}\delta^{(4)}(x-y). \tag{1.186}$$

Hence the generating functional can be written in the compact form

$$Z_0[\eta,\bar{\eta}] = Z_0[0,0]\,\exp\left[-\int \mathrm{d}^4x\,\mathrm{d}^4y\;\bar{\eta}(x)S_F(x-y)\eta(y)\right]. \tag{1.187}$$

In momentum space this propagator is

$$S_F(p) = \frac{\mathrm{i}}{\not{p} - m + \mathrm{i}\epsilon} = \frac{\mathrm{i}(\not{p}+m)}{p^2 - m^2 + \mathrm{i}\epsilon}. \tag{1.188}$$

Finally, the time-ordered two-point function is obtained by differentiating with respect to the Grassmann sources. With the derivative convention used above,

$$\langle 0|T\{\psi_a(x)\bar{\psi}_b(y)\}|0\rangle = -\frac{1}{Z_0[0,0]}\,\frac{\delta^2 Z_0[\eta,\bar{\eta}]}{\delta\bar{\eta}_a(x)\,\delta\eta_b(y)}\bigg|_{\eta=\bar{\eta}=0} = S_{F,ab}(x-y). \tag{1.189}$$

Thus the whole fermionic path integral is fixed by the same idea as the scalar Gaussian integral, but with Grassmann variables: the Gaussian gives a determinant, and insertions give the inverse Dirac operator, namely the fermion propagator.

5. Symmetries in Path Integral Formalism

The path integral approach provides a direct link between symmetries and physical observables through the following hierarchy: **Variational calculus** $\to$ **Schwinger-Dyson equations** $\to$ **Noether's theorem** $\to$ **Ward identities in QED**.

5.1 Schwinger-Dyson Equations

For simplicity, we still consider the scalar field theory as an example. By doing the partial integration, the action can be rewritten as:

$$S[\phi] = \int \mathrm{d}^4x\,\mathcal{L}[\phi] = \int \mathrm{d}^4x\left\{\frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2\right\} = -\frac{1}{2}\int \mathrm{d}^4x\,\phi\left(\Box + m^2\right)\phi. \tag{1.190}$$

The n-point correlator is given by:

$$\langle\Omega|T\{\phi(x_1)\ldots\phi(x_n)\}|\Omega\rangle = \frac{\int \mathcal{D}\phi\,\phi_1\ldots\phi_n e^{\mathrm{i}S[\phi]}}{\int \mathcal{D}\phi\,e^{\mathrm{i}S[\phi]}}. \tag{1.191}$$

We derive classical equations of motion from a variation of fields: $\phi(x) \to \phi'(x) = \phi(x) + \varepsilon(x)$, i.e., $\delta\phi(x) = \varepsilon(x)$. The integration measurement is invariant due to ε is infinitesimal: $\mathcal{D}\phi' = \mathcal{D}\phi$. And the action changes as: $S[\phi'] = S[\phi+\varepsilon] = S[\phi] + \delta S$. So we have

$$\int \mathcal{D}\phi\, \phi_1 \ldots \phi_n e^{\mathrm{i}S[\phi]} = \int \mathcal{D}\phi'\, \phi_1' \ldots \phi_n' e^{\mathrm{i}S[\phi+\varepsilon]} \tag{1.192}$$

$$= \int \mathcal{D}\phi\, e^{\mathrm{i}(S[\phi]+\delta S)} (\phi_1 + \varepsilon_1) \ldots (\phi_n + \varepsilon_n), \tag{1.193}$$

where we have denoted that $\phi_i \equiv \phi(x_i)$ and $\varepsilon_i \equiv \varepsilon(x_i)$. Then we expand ε to the first order, the exponential is $e^{\mathrm{i}(S+\delta S)} \simeq e^{\mathrm{i}S}(1 + \mathrm{i}\delta S)$. And for δS, we have

$$\delta S = \int \mathrm{d}^4 y \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu \delta\phi \right) \tag{1.194}$$

$$= \int \mathrm{d}^4 y \left[-m^2 \phi \varepsilon + \partial^\mu \phi \partial_\mu \varepsilon \right] \tag{1.195}$$

$$= - \int \mathrm{d}^4 y \left[\varepsilon(y) \left(\Box_y + m^2 \right) \phi(y) \right]. \tag{1.196}$$

Thus, put these result into (1.193), we have:

$$\int \mathcal{D}\phi\, e^{\mathrm{i}(S[\phi]+\delta S)} (\phi_1 + \varepsilon_1) \ldots (\phi_n + \varepsilon_n)$$

$$= \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[1 - \mathrm{i} \int \mathrm{d}^4 y\, \varepsilon(y) (\Box_y + m^2) \phi(y) \right] \left[(\phi_1 \ldots \phi_n) + (\varepsilon_1 \phi_2 \ldots \phi_n) + \cdots + (\phi_1 \ldots \varepsilon_n) \right] \tag{1.197}$$

$$= \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[(\phi_1 \ldots \phi_n) + (\varepsilon_1 \phi_2 \ldots \phi_n) + \cdots + (\phi_1 \ldots \varepsilon_n) \right]$$
$$- \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \int \mathrm{d}^4 y\, \mathrm{i} \left[\varepsilon(y) (\Box_y + m^2) \phi(y) \right] (\phi_1 \ldots \phi_n) + \mathcal{O}(\varepsilon^2). \tag{1.198}$$

Subtracting the first term $\int \mathcal{D}\phi\, \phi_1 \ldots \phi_n e^{\mathrm{i}S[\phi]}$ with (1.192), we have

$$0 = -\mathrm{i} \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left\{ \int \mathrm{d}^4 y\ \varepsilon(y) \left[(\Box_y + m^2) \phi(y) \right] \phi_1 \ldots \phi_n + \mathrm{i} \sum_{i=1}^{n} \phi_1 \ldots \epsilon_i \ldots \phi_n \right\}, \tag{1.199}$$

where ε_i will replace the corresponding ϕ_j when $i = j$ in the summation. And we can notice that $\varepsilon(x_i) = \int \mathrm{d}^4 y\ \varepsilon(y) \delta^{(4)}(x_i - y)$, so we have

$$\int \mathrm{d}^4 y\ \varepsilon(y) \left\{ -\mathrm{i} \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[\left[(\Box_y + m^2) \phi(y) \right] \phi_1 \ldots \phi_n + \mathrm{i} \sum_{i=1}^{n} \phi_1 \ldots \delta^{(4)}(y - x_i) \ldots \phi_n \right] \right\} = 0. \tag{1.200}$$

For arbitrary ε holds this relation, the only possible case is

$$\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[\left[(\Box_y + m^2) \phi(y) \right] \phi_1 \ldots \phi_n + \mathrm{i} \sum_{i=1}^{n} \phi_1 \ldots \delta^{(4)}(y - x_i) \ldots \phi_n \right] = 0. \tag{1.201}$$

Example: Consider two-point correlator, i.e., $n = 1$:

$$(\Box_y + m^2) \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \phi(y) \phi(x_1) = -\mathrm{i}\delta^{(4)}(y - x_1) \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} = -\mathrm{i}\delta^{(4)}(y - x_1) Z[J = 0] \tag{1.202}$$

Putting it back in the expression for the correlator:

$$(\Box_y + m^2)\underbrace{\langle\Omega|T\{\phi(y)\phi(x_1)\}|\Omega\rangle}_{G_F(y-x_1)} = -\mathrm{i}\delta^{(4)}(y-x_1). \tag{1.203}$$

Thus,

$$G_F(y-x_1) = -\mathrm{i}\delta^{(4)}(y-x_1)\left(\Box_y + m^2\right)^{-1}. \tag{1.204}$$

Similarly, we would like to generalize this form into n-point correlator. From (1.201) we have

$$\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}(\Box_y + m^2)\phi(y)\,(\phi_1\dots\phi_n) = -\mathrm{i}\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}\left(\sum_{i=1}^{n}\phi_1\dots\delta^{(4)}(y-x_i)\dots\phi_n\right). \tag{1.205}$$

Notice $-(\Box_y + m^2)\phi(y)$ in the left-hand side is nothing but the variation of the action:

$$\frac{\delta S[\phi]}{\delta\phi(y)} = \int \mathrm{d}^4x\left(\frac{\delta\mathcal{L}}{\delta\phi}\delta(x-y) + \frac{\delta\mathcal{L}}{\delta\partial_\mu\phi}\partial_\mu\delta(x-y)\right) = \frac{\delta\mathcal{L}}{\delta\phi(y)} - \partial_\mu\frac{\delta\mathcal{L}}{\delta\partial_\mu\phi(y)}, \tag{1.206}$$

the result of this variation always the equation of motion (or the Euler-Lagrange equation, if you like). So Without loss of generality, we can replace $-(\Box_y + m^2)\phi(y)$ by $\frac{\delta S[\phi]}{\delta\phi(y)}$ in (1.205) and divide $Z[0]$ as

$$\frac{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}\left(\frac{\delta S[\phi]}{\delta\phi(y)}\phi_1\dots\phi_n\right)}{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}} = \frac{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}\left(\mathrm{i}\sum_{i=1}^{n}\phi_1\dots\delta^{(4)}(y-x_i)\dots\phi_n\right)}{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}}. \tag{1.207}$$

We then get the **Schwinger-Dyson equation** for n-point functions:

$$\left\langle\Omega\left|\frac{\delta S[\phi]}{\delta\phi(y)}T\{\phi_1\dots\phi_n\}\right|\Omega\right\rangle = \mathrm{i}\sum_{i=1}^{n}\left\langle\Omega\left|T\{\phi_1\dots\delta^{(4)}(y-x_i)\dots\phi_n\}\right|\Omega\right\rangle. \tag{1.208}$$

We can simply denote it as:

$$\left\langle\frac{\delta S[\phi]}{\delta\phi(y)}T\{\phi_1\dots\phi_n\}\right\rangle = \mathrm{i}\sum_{i=1}^{n}\left\langle T\{\phi_1\dots\delta^{(4)}(y-x_i)\dots\phi_n\}\right\rangle. \tag{1.209}$$

5.2 Conservation Laws and Noether's Theorem

Consider a complex scalar field $\mathcal{L} = \partial_\mu\phi\partial^\mu\phi^* - m^2\phi\phi^*$. Under the *global* $U(1)$ transformation $\phi \to \phi' = e^{\mathrm{i}\alpha}\phi$, the variation is:

$$\delta\phi = \mathrm{i}\alpha\phi; \qquad \text{for } \alpha\in\mathbb{R}, \tag{1.210}$$

with the unchanged Lagrangian. For a *local* transformation $\alpha \to \alpha(x)$, we have $\delta\phi(x) = \mathrm{i}\alpha(x)\phi(x)$, so the variation for the Lagrangian is

$$\begin{aligned}\delta\mathcal{L} &= \mathcal{L}[\phi'] - \mathcal{L}[\phi]\\ &= \partial_\mu\phi\partial^\mu\phi^* + \mathrm{i}\partial_\mu\alpha(x)\left(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi\right) + \mathcal{O}(\alpha^2) - m^2\phi\phi^* - \partial_\mu\phi\partial^\mu\phi^* + m^2\phi\phi^*\\ &= \partial_\mu\alpha(x)j^\mu(x),\end{aligned} \tag{1.211}$$

where $j^\mu(x) = \mathrm{i}\left(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi\right)$. An n-point correlator is invariant under this transformation

$$\int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}(\phi_1\dots\phi_{n/2})(\phi_1^*\dots\phi_{n/2}^*) = \int \mathcal{D}\phi'\mathcal{D}\phi'^*\, e^{\mathrm{i}S'[\phi',\phi'^*]}(\phi_1'\dots\phi_{n/2}')(\phi_1'^*\dots\phi_{n/2}'^*), \tag{1.212}$$

Notice that $e^{\mathrm{i}S'[\phi',\phi'^*]} = e^{\mathrm{i}(S+\delta S)} \simeq e^{\mathrm{i}S}(1+\mathrm{i}\delta S)$. Expand the right-hand side to order $\alpha(x)$, and subtracting with the left-hand side, we have

$$0 = \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\,(1+\mathrm{i}\delta S)\,((1+\mathrm{i}\alpha)\phi_1\dots(1+\mathrm{i}\alpha)\phi_n^*) \tag{1.213}$$

$$= \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\Big\{\mathrm{i}\delta S(\phi_1\dots\phi_n^*) + [(\mathrm{i}\alpha\phi_1)\dots\phi_n^*] + \dots + [\phi_1\cdots(\pm\mathrm{i}\alpha\phi_i^{(*)})\dots\phi_n^*] + \dots + [\phi_1\dots(-\mathrm{i}\alpha\phi_n^*)]\Big\} \tag{1.214}$$

$$= \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{\mathrm{i}\int \mathrm{d}^4y\,\partial_\mu\alpha(y)j^\mu(y)(\phi_1\dots\phi_n^*) + \sum_{i=1}^n \phi_1\cdots(\pm\mathrm{i}\alpha\phi_i^{(*)})\dots\phi_n^*\right\}, \tag{1.215}$$

where $\pm\mathrm{i}\phi^{(*)_i}$ denote that for ϕ_i^* taking $-\mathrm{i}$ and for ϕ_i taking $+\mathrm{i}$ .After partial integration to factor out $\alpha(y)$:

$$0 = \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{\int \mathrm{d}^4y\,\alpha(y)\partial_\mu j^\mu(y)\phi_1\dots\phi_n + \mathrm{i}\sum_{i=1}^n \phi_1\dots(\pm\mathrm{i})\left[\int \mathrm{d}^4y\,\delta(y-x_i)\alpha(y)\right]\phi_i^{(*)}\dots\phi_n\right\} \tag{1.216}$$

$$= \int \mathrm{d}^4y\,\alpha(y)\int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{-\mathrm{i}\partial_\mu j^\mu(y)\phi_1\dots\phi_n + \mathrm{i}\sum_{i=1}^n \phi_1\dots(\pm\mathrm{i})\delta(y-x_i)\phi_i^{(*)}\dots\phi_n\right\}. \tag{1.217}$$

Similarly, for any infinitesimally α holding this relation, so the only possible way is

$$\int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{-\mathrm{i}\partial_\mu j^\mu(y)\phi_1\dots\phi_n + \mathrm{i}\sum_{i=1}^n \phi_1\dots(\pm\mathrm{i})\delta(y-x_i)\phi_i^{(*)}\dots\phi_n\right\} = 0. \tag{1.218}$$

We can also use the correlator notation:

$$\langle\partial_\mu j^\mu(y)\phi(x_1)\dots\phi(x_n)\rangle = -\mathrm{i}\sum_{i=1}^n\langle\phi(x_1)\dots(\pm\mathrm{i})\delta(y-x_i)\phi_i^{(*)}\dots\phi(x_n)\rangle. \tag{1.219}$$

Now we generalize this relation to a set of real fields ϕ_a. For a local transformation $\phi_a(x)\to\phi'_a(x) = \phi_a(x)+\varepsilon\Delta\phi_a(x)$, if the global transformation gives $\mathcal{L}\to\mathcal{L}+\varepsilon\partial_\mu J^\mu$, then for a local $\varepsilon(x)$ we have

$$\partial_\mu\phi'_a = \partial_\mu(\phi_a+\varepsilon\Delta\phi_a) = \partial_\mu\phi_a + \varepsilon\partial_\mu(\Delta\phi_a) + (\partial_\mu\varepsilon)\Delta\phi_a. \tag{1.220}$$

So when we expand the Lagrangian, we have:

$$\mathcal{L}(\phi',\partial\phi') = \mathcal{L}(\phi,\partial\phi) + \sum_a\left(\frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\delta(\partial_\mu\phi_a)\right), \tag{1.221}$$

then plug the variation $\delta\phi_a = \varepsilon\Delta\phi_a$ into it and using the Euler-Lagrange equation:

$$\delta\mathcal{L} = \sum_a\left[\frac{\partial\mathcal{L}}{\partial\phi_a}(\epsilon\Delta\phi_a) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}(\epsilon\partial_\mu(\Delta\phi_a)+(\partial_\mu\epsilon)\Delta\phi_a)\right] \tag{1.222}$$

$$= \sum_a\left[\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}(\epsilon\Delta\phi_a) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}(\epsilon\partial_\mu(\Delta\phi_a)+(\partial_\mu\epsilon)\Delta\phi_a)\right] \tag{1.223}$$

$$= \sum_a\left[\varepsilon\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\Delta\phi_a\right) + (\partial_\mu\epsilon)\Delta\phi_a\right]. \tag{1.224}$$

Then we get the variation for the Lagrangian:

$$\mathcal{L}(\phi_a) \to \mathcal{L}(\phi_a') = \mathcal{L}(\phi_a) + \varepsilon(x)\partial_\mu J^\mu + \partial_\mu \varepsilon(x) \sum_a \Delta\phi_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \tag{1.225}$$

$$= \mathcal{L}(\phi_a) + \varepsilon(x)\partial_\mu J^\mu - \varepsilon(x)\partial_\mu \left(\sum_a \Delta\phi_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \tag{1.226}$$

$$= \mathcal{L}(\phi_a) - \varepsilon(x)\partial_\mu \left(\sum_a \Delta\phi_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} - J^\mu \right), \tag{1.227}$$

which appears an extra term due to the infinitesimal transformation rely on the coordinate. Therefore, the Noether current j^μ can be defined from the variation for the action

$$S[\phi_a'] = \int \mathrm{d}^4 y \, [\mathcal{L}[\phi_a] - \varepsilon(y)\partial_\mu j^\mu(y)] = S[\phi_a] - \int \mathrm{d}^4 y \, \varepsilon(y)\partial_\mu j^\mu(y). \tag{1.228}$$

Thus, the Noether current j^μ is

$$j^\mu = \sum_a \Delta\phi_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} - J^\mu. \tag{1.229}$$

Put this current back to (1.219), we get the conserved symmetry Schwinger-Dyson equation as:

$$\langle \partial_\mu j^\mu(y) \phi_{a_1}(x_1) \ldots \phi_{a_n}(x_n) \rangle = -\mathrm{i} \sum_{i=1}^{n} \langle \phi_{a_1}(x_1) \ldots \delta(y - x_i) \Delta\phi_{a_i} \ldots \phi_{a_n}(x_n) \rangle \tag{1.230}$$

5.3 Ward-Takahashi Identity in QED

Consider the QED Lagrangian with fermions:

$$\mathcal{L}[\psi, \bar{\psi}, A_\mu] = \bar{\psi}(\mathrm{i}\gamma^\mu D_\mu - m)\psi + A_\mu j^\mu, \tag{1.231}$$

where $j^\mu = e\bar{\psi}\gamma^\mu\psi$. The global $U(1)$ transformation gives

$$\psi \to e^{\mathrm{i}\alpha}\psi \implies \delta\psi = \mathrm{i}\alpha\psi. \tag{1.232}$$

So the local transformation gives

$$\mathcal{L}[\psi', \bar{\psi}'] = \mathcal{L}[\psi, \bar{\psi}] - (\partial_\mu \alpha) j^\mu. \tag{1.233}$$

Applying Schwinger-Dyson equation with two fields, we have

$$\partial_\mu \langle 0|T\{j^\mu(y)\psi(x_1)\bar{\psi}(x_2)\}|0\rangle = \delta^{(4)}(y-x_1)\langle 0|T\{\psi(x_1)\bar{\psi}(x_2)\}|0\rangle - \delta^{(4)}(y-x_2)\langle 0|T\{\psi(x_1)\bar{\psi}(x_2)\}|0\rangle. \tag{1.234}$$

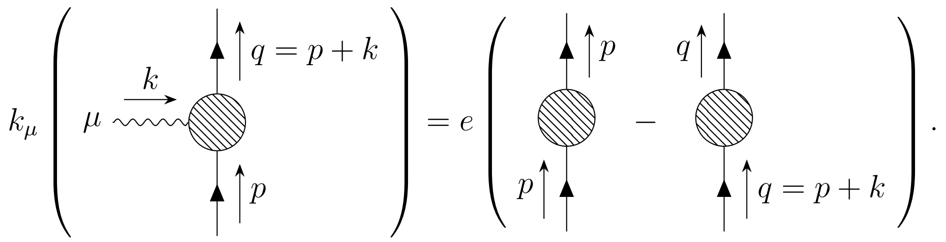
The minus sign shows that ψ follows **Fermi statistics**. Take a Fourier transform to momentum space:

$$k_\mu \langle 0|T\{j^\mu(k)\psi(q)\bar{\psi}(p)\}|0\rangle = e \left(\langle 0|T\{\psi(q-k)\bar{\psi}(p)\}|0\rangle - \langle 0|T\{\psi(q)\bar{\psi}(p+k)\}|0\rangle \right), \tag{1.235}$$

which is nothing but the **Ward-Takahashi identity** in QED that we are familiar with:

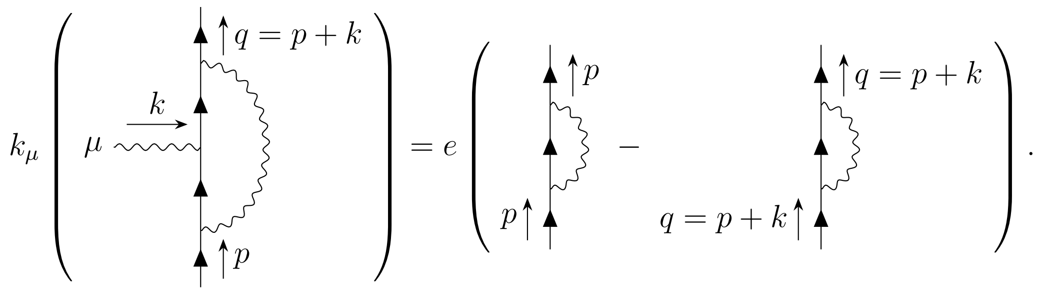
$$k_\mu \mathcal{M}^\mu(k; p, q = p + k) = e\mathcal{M}_0(p) - e\mathcal{M}_0(q = p + k). \tag{1.236}$$

It can also be expressed by the Feynman diagram. Notice that the interaction $e\bar{\psi}\gamma^\mu\psi A_\mu$ contains one ψ and one $\bar{\psi}$, but this does not mean that the two fermion segments in an electron scattering diagram should be drawn as two disconnected particles. They are the two ends of one continuous fermion line. With the photon momentum k flowing into the current insertion, momentum conservation gives $q = p + k$:



(1.237)

Especially for the one-loop vertex correction, the same convention must be used. The fermion line is continuous; the upper external momentum is therefore $q = p + k$, not $p - k$:



(1.238)

The lower and upper segments in the first diagram are not separate particle species. They are the incoming and outgoing parts of the same fermion line. Algebraically this corresponds to the two fields $\bar{\psi}$ and ψ at the QED vertex; graphically it is represented by one continuous fermion arrow.